

Atharva Mavale

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CAREER PROFILE

I am a graduate AI engineer with strong foundations in machine learning, computer vision, and full-stack web development. Experienced in building and deploying LLM-powered applications using Fast API, React, and OpenAI APIs. Passionate about AI for education, Socratic tutoring systems, and intelligent interfaces that improve learning outcomes. I am currently completing Master of Applied AI at Deakin University (Melbourne) and am actively seeking opportunities in AI engineering.

EDUCATION

Pillai College of Arts Commerce and Science - Bachelor of Science in information technology (BSc IT) | 2020-2023

Grade with an 8.38 CGPA

Deakin University (Australia) - Master of Applied Artificial Intelligence (Professional) | 2024 – 2026 (Ongoing)

SKILLS

Languages: Python (Advanced), Java, R, SQL

AI/ML Libraries: OpenCV, NLTK, LangChain, Hugging Face Transformers, scikit-learn, PyTorch

Computer Vision: Hand gesture recognition, real-time object detection, video processing, frame analysis

Data Analysis: Excel (Pivot Tables, statistical analysis), Tableau, data visualization

Technical Skills: Data Structures, OOP, Machine Learning, NLP, ROS, YOLO

PROJECTS AND ASSESSMENTS

StudyMate AI – AI Socratic Tutor for Code Repositories

Technologies: Fast API, Streamlit, OpenAI GPT-4o-mini, LangChain, Python | Render, Streamlit Cloud

- Extended the Socratic tutoring concept into an agentic, repo-aware system capable of adapting explanations to any codebase.
- Implemented session-based multi-turn conversations with adaptive teaching behavior across beginner, intermediate, and advanced knowledge levels.
- Deployed full-stack app with FastAPI backend on Render and Streamlit UI on Streamlit Community Cloud with health-check endpoints and Swagger API docs.
- **Submitted to Sophie AI Agents Hackathon (Feb 2026)** – a competitive 33-hour solo hackathon open to Australian developers and university students, with a \$5,000 prize and job offer opportunities for top performers.
- [GitHub Repository](#)

Project Socrates Developer – AI Socratic Tutor

Technologies: Fast API, React, Vite, OpenAI GPT-4.1-mini, Render, Vercel

- Built a full-stack web application providing Socratic, question-driven tutoring using GPT-4.1-mini via a Fast API backend and React/Vite chat interface.
- Implemented custom NLP preprocessing (regex-based tokenization, simple lemmatization) to analyze user input and return sentence/word counts alongside AI responses.
- Integrated error handling for LLM/network failures and deployed backend to Render and frontend to Vercel with environment-based configuration.
- Live demo: [Vercel URL](#) | Code: [GitHub link](#)

Autonomous Mine Safety RAG Copilot (Maintenance + Safety + PPE Vision)

Technologies: Fast API, React, Vite, Python, RAG, YOLO PPE detection (Still in progress)

- Built a RAG copilot that answers mining maintenance/safety questions with source-linked citations from manuals, standards, incidents, and site procedures.
- Fixed “info scattered + keyword search fails” by enabling natural-language Q&A that retrieves the right passages (e.g., wording mismatch cases)
- implemented an end-to-end RAG pipeline: index documents, retrieve relevant chunks, and generate short answers that cite the exact sources used.

- Currently expanding the PDF corpus and improving retrieval via rebuild → query testing → chunking refinements.

[GitHub Repository](#)

Math with Computer Vision

Technologies: Python, OpenCV, Gesture Recognition

- Developed an AI-powered tool that recognizes hand-drawn equations in the air using OpenCV and gesture recognition.
- Enabled real-time solving of math problems from algebra to calculus through computer vision techniques.
- Delivered an intuitive, hands-free experience for math interaction, tested with multiple sample equations.
- Reduced average solution time compared to manual input by approximately 60%.
- Helped kickstart learning in deep learning and gesture-based recognition systems.

[GitHub Repository](#)

PROFESSIONAL EXPERIENCE

AI Assisted Navigation Device

InnovAlte , Australia

- Collaborating in a capstone project that aims to equip individuals with disabilities with an AI-powered navigation solution.
 - Contributing to computer vision module for real-time object recognition with planned integration of YOLO detection algorithm
 - Participating in bi-weekly sprint planning sessions and weekly team standups following Agile development methodology
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CURRENT INVOLVEMENT

Deakin Competitive Robotics ([Link](#))

- Developing ROS (Robot Operating System) skills to enhance robot automation and navigation capabilities.
 - Collaborated on recruitment and onboarding activities for team members during competitive robotics preparation.
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CERTIFICATIONS

Python | Techgen Computer Training Institute, Navi Mumbai | 25 Sep 2023 – 05 Dec 2023

- Completed 60+ hour intensive Python course covering 8 core topics: OOP, decorators, generators, functional programming, error handling, data structures, file I/O, and debugging
- Developed 5+ hands-on projects applying Python to data analysis, automation scripts, and algorithm implementation
- Applied course learnings directly to AI projects including computer vision applications and data preprocessing pipelines

[Course Portfolio](#)

After Effects | Frameboxx Vashi - Gaming, Animation & VFX Training Institute, Navi Mumbai | 22 March 2024 – 10 May 2024

- Completed 40-hour After Effects course covering 2D/3D tracking, compositing, keyframing, and visual effects workflows
- Created 3 portfolio projects demonstrating motion design, video compositing, and visual storytelling techniques
- Developed foundation in animation supporting future AI-driven multimedia and computer vision visualization projects

[Portfolio Folder](#)

PROFESSIONAL SKILLS

- Remote collaboration and communication
- Agile/SCRUM workflows
- Technical documentation